Identifier Validation with Reasoning

- **Q1**
    - **Input:** my_var = 4
    - **Output:** True
    - **Reason:** It starts with a letter and contains only valid alphanumeric characters and underscores.
- **Q2**
    - **Input:** 2value = 4
    - **Output:** False
    - **Explanation:** Identifiers cannot start with a number.
    - **Corrected:** value2 = 4
- **Q3**
    - **Input:** user_name = 4
    - **Output:** True
    - **Reason:** It complies with Python naming rules by using valid characters and snake_case formatting.
- **Q4**
    - **Input:** user-name = 4
    - **Output:** False
    - **Explanation:** Identifiers cannot contain hyphens (-)
    - **Corrected:** user_name = 4
- **Q5**
    - **Input:** _count = 4
    - **Output:** True
    - **Reason:** It is valid for an identifier to start with an underscore (_).
- **Q6**
    - **Input:** class = 4
    - **Output:** False

    - **Explanation:** class is a reserved keyword in Python used to define classes and cannot be used as an identifier.
    - **Corrected:** class_name = 4
- **Q7**
    - **Input:** totalAmount = 4
    - **Output:** True
- **Q8**
    - **Input:** first name = 4
    - **Output:** False
    - **Explanation:** Identifiers cannot contain spaces.
    - **Corrected:** first_name = 4
- **Q9**
    - **Input:** data123 = 4
    - **Output:** True
    - **Reason:** It starts with letters and contains numbers afterwards, which is allowed.
- **Q10**
    - **Input:** @value = 4
    - **Output:** False
    - **Explanation:** Identifiers cannot contain special characters like @.
    - **Corrected:** value = 4
- **Q11**
    - **Input:** True = 4
    - **Output:** False
    - **Explanation:** True is a built-in Boolean values/literal and cannot be reassigned as a variable name.
    - **Corrected:** is_true = 4
- **Q12**
    - **Input:** value_1 = 4

    - **Output:** True
    - **Reason:** It starts with a letter and properly combines letters, underscores, and numbers.
- **Q13**
    - **Input:** def = 4
    - **Output:** False
    - **Explanation:** def is a reserved keyword in Python used to define functions.
    - **Corrected:** def_val = 4
- **Q14**
    - **Input:** init = 4
    - **Output:** True
    - **Reason:** init is a normal word and is not a reserved Python keyword (unlike the double-underscore magic method __init__).
- **Q15**
    - **Input:** user$ = 4
    - **Output:** False
    - **Explanation:** Identifiers cannot contain the special character $.
    - **Corrected:** user_currency = 4
- **Q16**
    - **Input:** Name = 4
    - **Output:** True
    - **Reason:** It starts with an uppercase letter, which is completely valid in Python.
- **Q17**
    - **Input:** value! = 4
    - **Output:** False
    - **Explanation:** Identifiers cannot contain the special character !.
    - **Corrected:** value = 4
- **Q18**
    - **Input:** a123 = 4

    - **Output:** True
    - **Reason:** It starts with a valid letter followed by valid numbers.
- **Q19**
    - **Input:** while = 4
    - **Output:** False
    - **Explanation:** while is a reserved keyword used to construct loops.
    - **Corrected:** while_loop = 4
- **Q20**
    - **Input:** snake_case_variable = 4
    - **Output:** True
    - **Reason:** It follows perfectly valid snake_case rules using only lowercase letters and underscores.